

Ahmad Sebaq

Phone: +201092003631
Email: ahmad.sebaq@gmail.com
LinkedIn: [ahmad-sebaq-807273104](https://www.linkedin.com/in/ahmad-sebaq-807273104)
GitHub: github.com/asebaq

SUMMARY

Senior Computer Vision & AI engineer (7 years) specializing in generative models, vision-language models, and agentic LLM pipelines. Technical lead who has shipped research to production and led teams of up to 8.

EDUCATION

Nile University M.S. in Communication and Information Technology, GPA: 4.0/4.0	Giza, Egypt 2021–2026
Minia University B.S. in Biomedical Engineering, Grade: Very Good with Honors (84.6%)	Minya, Egypt 2014–2019

EXPERIENCE

CyShield Senior Computer Vision Engineer (<i>Full-time</i>)	New Cairo, Egypt January 2025 – Current
– Biometric Border Control: Built face- and fingerprint-based identity verification for border-control checkpoints — face detection, quality/liveness filtering, embedding extraction and fingerprint minutiae matching, with 1:1 verification and 1:N identification against watchlist galleries under strict latency and false-match constraints. – Stadium Video Surveillance: Delivered a multi-camera stadium surveillance system — person detection, multi-object tracking and re-identification across overlapping camera views, plus action recognition for crowd-incident and prohibited-behaviour alerts on live video streams. – Generative Models: Designed and trained diffusion- and GAN-based inpainting models for image restoration and content-aware editing, owning the full pipeline: dataset curation, mask generation, architecture selection, training at scale, and evaluation against perceptual (LPIPS, FID) and structural (SSIM, PSNR) metrics. – VLM Fine-tuning: Fine-tuned vision-language models on domain-specific image-text pairs to improve grounding, captioning, and visual question-answering accuracy on in-house data. Built semantic search and retrieval pipelines over vector databases (embeddings indexing, similarity search, hybrid filters) to power RAG and visual retrieval workflows. – Technical Leadership: Led a squad of 3 AI engineers — scoping work, code review, mentoring on generative-model best practices, and aligning research direction with product deadlines.	
CNTEXT AI/Computer Vision Consultant (<i>Part-time</i>)	Remote — UAE / Egypt January 2026 – Current
– Robotics Data & Teleoperation: Built an egocentric data-collection and QA platform processing 100+ hours of 3-camera teleoperation video per week into validated LeRobot datasets for imitation learning — 3D hand-pose reconstruction, automated video QA, and VLM-based action annotation. – Speech & Multi-modal CV: Designed a multi-stage Arabic ASR pipeline (denoising, diarization, dialect-aware transcription) and multi-modal CV systems for instance segmentation and cross-modal RGB/IR drone-image alignment. – Agentic RAG (Arabic): Shipped a production Arabic retrieval service on LangGraph — intent router plus an evaluator-optimizer loop (multi-query fan-out, RRF hybrid dense/BM25 search, cross-encoder reranking, groundedness reflection) answering with streamed inline citations. Lifted Hit@1 0.84 → 0.95, nDCG@10 0.85 → 0.93, and correct “no-answer” refusal 55% → 94% on a 278-question LLM-judged benchmark. <i>Stack:</i> FastAPI, Qdrant, GPT-4o, Qwen3 embed/rerank, Postgres, Docker.	

Lumin Soft

Giza, Egypt

AI Team Lead (*Full-time*)

November 2023 – January 2025

- **Team Leadership:** Led a team of 8 AI developers across segmentation, detection, tracking, and OCR projects — running 1:1s, sprint planning, and technical mentorship.
- **AI Solution Development:** Delivered 3 production projects spanning segmentation, detection, tracking, clustering, re-identification (ReID), and OCR.
- **Sports Analytics (flagship):** Led a multi-model sports video pipeline — player/ball detection & tracking, team clustering, ReID, action recognition, possession tracking — output as annotated video plus metrics reports.
- **Research-to-Production:** Turned research prototypes into deadline-bound production pipelines.

Applied Innovation Center

Giza, Egypt

R&D Engineer (*Full-time*)

October 2022 – November 2023

- **Land Cover Detection:** Segmented land cover into 7 classes from Sentinel-2 and Planet satellite imagery using U-Net and Transformer networks, reaching 98% overall accuracy across Egypt's main governorates.
- **Geospatial Pipelines:** Built end-to-end satellite-imagery pipelines (preprocessing, tiling, model inference, post-processing), collaborating with GIS analysts on validation and deployment.

Center for Informatics Science, Nile University

Giza, Egypt

Research Assistant (*Full-time*)

April 2021 – April 2023

- **Disease Classification:** Classified rice crop diseases across 5 classes with ResNet and Transformer models, achieving 96% F1 on a 250K-image dataset.
- **Kaggle Competitions:** Participated in computer vision challenges, including UW-Madison Gastrointestinal Tract image segmentation.

Dawarha

Cairo, Egypt

Computer Vision Engineer (*Part-time*)

April 2021 – September 2023

- Built a real-time YOLOv3 + KMeans pipeline for 2-class recyclable-material detection (90 mAP), [deployed on a Raspberry Pi at 11 FPS](#).

VLens

Cairo, Egypt

Computer Vision Engineer (*Part-time*)

February 2021 – November 2021

- Built national-ID OCR with U-Net text segmentation and KMeans preprocessing, [deployed on AWS EC2](#).

Affectiva

Massachusetts, USA

Mentor (*Contract, 3 months*)

June 2020 – August 2020

- Mentored student teams through CV project scoping, model selection, and demo delivery.

TechnoIsland

Cairo, Egypt

Computer Vision Engineer (*Part-time*)

October 2019 – November 2021

- Built a CV pipeline for sperm-sample analysis — YOLOv3 detection, Yolact segmentation, DeepSort tracking across video frames.

PROJECTS

BlacksmithAI (*Freelance, 2026*)

blacksmithai.org

Natural-language to CAD agentic pipeline

- **Agentic pipeline:** Designed a multi-stage state machine (decompose → plan → assemble → execute → validate → repair) over a deterministic JSON IR with content-addressed caching, multi-modal image grounding, and human-in-the-loop review.
- **LLM tool-use & verification:** Built a Pydantic-validated tool catalog and pure verifier tools the LLM uses to self-check IR before execution; routed Claude Haiku and Gemini Flash via OpenRouter. *Stack:* FastAPI, Qdrant + BGE RAG, CadQuery, React + react-three-fiber, Docker.

TV Ad-Recognition Pipeline

Real-time broadcast ad detection via multi-modal fingerprinting

- Built a hybrid recognition system matching live TV broadcasts against an ad catalog using audio fingerprinting (Chromaprint), speaker embeddings (WavLM x-vectors), and scene-change keyframes (PySceneDetect + pHash + CLIP ViT-B/32 embeddings), fused over Elasticsearch kNN + BM25 search with NMS-suppressed airing events. Added a persistent-socket WebSocket endpoint for live per-channel recognition. *Stack:* two FastAPI services, Elasticsearch, PyTorch, FFmpeg; deployed as auto-start Windows services.

Sperm Analysis (CASA)

Computer-aided semen analysis desktop product

- **Analysis engine:** Built a CASA pipeline measuring sperm count, motility (per-cell kinematics via multi-object tracking), and morphology from microscopy and industrial-camera video — YOLO detection served through ONNX Runtime.
- **Product & deployment:** Shipped as an offline Windows desktop app — FastAPI backend (PyInstaller) wrapped in an Electron + React UI, industrial-camera capture (GenTL / Harvesters), encrypted model weights, HTTPS licensing service, and PDF clinical reports.

RESEARCH

Fatima Fellowship

Research Intern (*Part-time*)

Stanford, USA

September 2023 – August 2024

- Developed generative models to enhance low-resolution MRI images, focusing on model optimization and validation for clinical applications.

Linköping University

Research Intern (*Part-time*)

Linköping, Sweden

October 2022 – January 2023

- Developed GANs to manipulate Maxar aerial images by adding realistic seasonal effects, collaborating with geospatial analysts to ensure accuracy.

PUBLICATIONS

- [1] T. A. Abdel Salam, A. M. Abdelsalam, N. R. Sholkamy, and **Sebaq, Ahmad**, “Assessment of artificial intelligence in detecting errors on panoramic radiographs”, *Imaging Science in Dentistry*, vol. 56, e12, 2026. DOI: [10.5624/isd.20250242](https://doi.org/10.5624/isd.20250242)
- [2] **Sebaq, Ahmad** and M. ElHelw, “Rsdiff: Remote sensing image generation from text using diffusion model”, *Neural Computing and Applications*, vol. 36, no. 36, pp. 23 103–23 111, 2024. DOI: [10.1007/s00521-024-10363-3](https://doi.org/10.1007/s00521-024-10363-3)
- [3] Y. A. Alnaggar, **Sebaq, Ahmad**, K. Amer, E. Naeem, and M. Elhelw, “Rice plant disease detection and diagnosis using deep convolutional neural networks and multispectral imaging”, in *International Conference on Model and Data Engineering*, Springer, 2023, pp. 16–25. DOI: [10.1007/978-3-031-21595-7_2](https://doi.org/10.1007/978-3-031-21595-7_2)
- [4] M. S. M. Bahgat and **Sebaq, A. M.**, “An Analytical Computational Algorithm for Solving a System of Multipantograph DDEs Using Laplace Variational Iteration Algorithm”, in *Advances in Astronomy*, Jun. 2021, p. 16. DOI: [10.1155/2021/7741166](https://doi.org/10.1155/2021/7741166)

AWARDS

- Full Master Scholarship from Nile University 2021–2023
- 3rd Place, Global Monitoring for Environment and Security (GMES) & Africa Incubation Challenge 2021
- 3rd Place in Affectiva Makeathon with Inside Out project 2020
- Nokia Africathon Finalist with ALS project 2019

SKILLS

- **Programming:** Python, C++, JavaScript / TypeScript, React
- **Generative AI & LLMs:** LLM orchestration (LangGraph, LangChain), multi-agent pipelines, tool-use / function-calling, agentic RAG, multi-modal VLMs, local inference (llama-cpp-python), vector & hybrid search (Qdrant, Elasticsearch; dense + BM25, RRF fusion, cross-encoder reranking, ColBERT late interaction), LLM-as-judge evaluation, prompt engineering
- **Computer Vision:** OpenCV; segmentation (U-Net, Mask R-CNN, Swin), detection & tracking (YOLO, Yolact, DeepSort), OCR, re-identification, 3D hand-pose / SLAM, CLIP cross-modal matching
- **ML / Deep Learning:** PyTorch, ONNX Runtime, HuggingFace, scikit-learn; CNNs, Transformers, diffusion models, GANs; ASR (Whisper), speaker diarization (Pyannote), denoising (DeepFilterNet3), audio fingerprinting
- **MLOps & Deployment:** FastAPI, Docker, Jenkins CI, AWS (EC2), Git; Postgres + SQLAlchemy/Alembic, JWT auth, structured logging & Sentry; desktop packaging (Electron, PyInstaller); edge deployment (Jetson Nano & Orin, Raspberry Pi)
- **Robotics & Geospatial:** LeRobot imitation-learning data pipelines, industrial-camera capture; QGIS, Sentinel-2 / Planet / Maxar satellite imagery